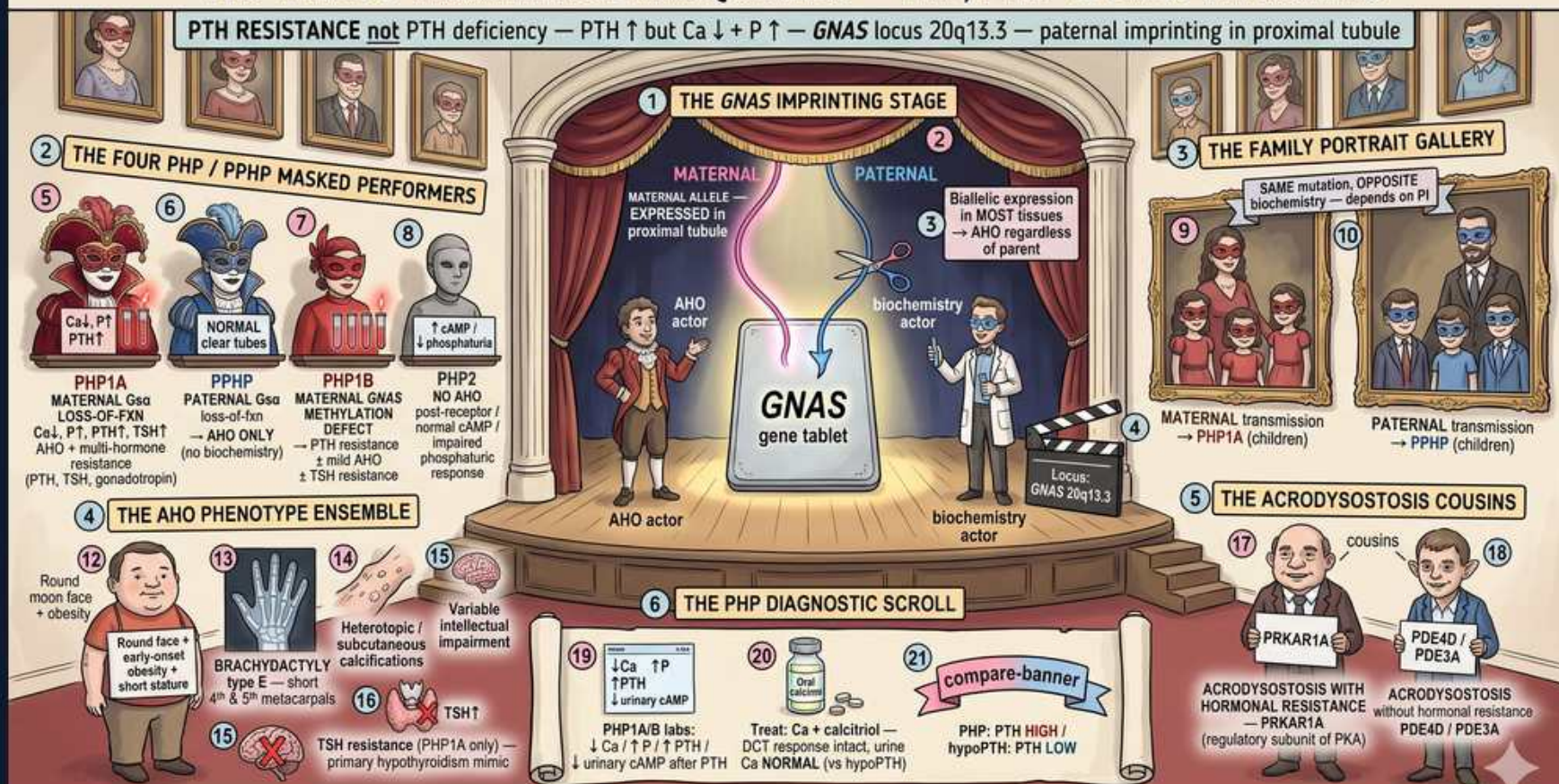


# Parathyroid/Bone — Pseudohypoparathyroidism (GNAS Masquerade)

## THE PSEUDO-PARATHYROID MASQUERADE — PHP, PPHP & GNAS IMPRINTING

PTH RESISTANCE not PTH deficiency — PTH ↑ but Ca ↓ + P ↑ — *GNAS* locus 20q13.3 — paternal imprinting in proximal tubule



PHP = PTH RESISTANCE, not deficiency. SAME mutation, OPPOSITE biochemistry — depends on parent. AHO regardless. PHP1A also resists TSH/gonadotropins. Distal tubule is with treatment.



## 1. PTH resistance, not deficiency

### WHAT YOU SEE

Subtitle: PTH RESISTANCE not PTH deficiency — PTH HIGH but Ca low + P high; GNAS 20q13.3

### WHAT IT ENCODES

Pseudohypoparathyroidism (PHP) is END-ORGAN RESISTANCE to PTH, not a deficiency of it. Target tissues (kidney, bone) can't respond, so the biochemistry MIMICS hypoparathyroidism — LOW calcium, HIGH phosphate — but the parathyroids sense the low Ca and crank out PTH, so PTH is HIGH (and 25-OH-D is normal). The defect is in GNAS on chromosome 20q13.3 (Gs-alpha of the PTH-receptor signaling cascade).

### BOARD TRAP

WRONG: calling low Ca + high phosphate 'hypoparathyroidism' without checking the PTH. True hypoPTH has a LOW/inappropriately-normal PTH; pseudohypoPTH has a HIGH PTH (resistance). The discriminator is the PTH level: low Ca + high PO4 + HIGH PTH (with normal renal function and vitamin D) = PTH resistance = pseudohypoPTH.

## 2. GNAS gene / Gs-alpha

### WHAT YOU SEE

Center stage: GNAS gene tablet, the Gs-alpha subunit defect on chromosome 20q13.3

### WHAT IT ENCODES

GNAS (20q13.3) encodes the alpha subunit of the stimulatory G protein (Gs-alpha) that couples the PTH receptor to adenylyl cyclase. A LOSS-OF-FUNCTION mutation means PTH binds its receptor but can't generate cAMP -> no downstream effect -> functional PTH resistance. Because Gs-alpha also transduces TSH, gonadotropin, GHRH and other receptors, the resistance can be MULTI-HORMONAL.

### BOARD TRAP

WRONG: assuming the parathyroid GLAND or the PTH RECEPTOR itself is defective. The lesion is downstream — in the Gs-alpha signal transducer (GNAS). That's why one mutation produces resistance to several different hormones (PTH, TSH, gonadotropins): they all share the Gs-alpha/cAMP pathway.

## 3. Imprinting / maternal allele

### WHAT YOU SEE

GNAS Imprinting Stage: maternal allele expressed in proximal tubule; paternal imprinting

### WHAT IT ENCODES

GNAS is IMPRINTED in a tissue-specific way: in the renal proximal tubule (and thyroid, gonad, pituitary) only the MATERNAL allele is expressed; the paternal allele is silenced. So if the loss-of-function mutation is inherited from the MOTHER, the active renal allele is the broken one -> renal PTH resistance (hormone resistance). If inherited from the FATHER, the silenced paternal copy is mutated and the active maternal copy still works in kidney -> NO hormone resistance.

### BOARD TRAP

WRONG: predicting biochemistry from the mutation alone, ignoring parent of origin. The SAME GNAS loss-of-function mutation gives OPPOSITE hormonal phenotypes depending on which parent transmitted it — maternal = resistance (PHP1A), paternal = no resistance (PPHP). Imprinting is the whole point.

## 4. PHP1A (maternal)

### WHAT YOU SEE

Masked performer PHP1A: maternal GNAS loss-of-fxn; Ca low / PTH high; multi-hormone resistance + AHO

### WHAT IT ENCODES

PHP type 1A: MATERNALLY-inherited GNAS loss-of-function. Full phenotype = biochemical RESISTANCE (low Ca, high PO4, HIGH PTH) PLUS resistance to other Gs-alpha hormones (TSH -> hypothyroidism, gonadotropins -> hypogonadism, GHRH -> short stature) PLUS the Albright hereditary osteodystrophy (AHO) body habitus.

### BOARD TRAP

WRONG: expecting only calcium problems in PHP1A. It is the MULTI-hormone-resistance subtype WITH the AHO physical phenotype — so a child with brachydactyly, round face, short stature, AND hypocalcemia AND a high TSH fits PHP1A. Missing the associated hypothyroidism/hypogonadism is a classic oversight.

## 5. PseudoPHP (paternal)

### WHAT YOU SEE

Masked performer PPHP: paternal GNAS loss-of-fxn; NORMAL calcium/biochemistry

### WHAT IT ENCODES

Pseudo-PSEUDO-hypoparathyroidism (PPHP): PATERNALLY-inherited GNAS loss-of-function. Because the active maternal renal allele is intact, there is NO hormone resistance -> NORMAL calcium, normal phosphate, normal PTH. The patient has the AHO PHENOTYPE only (the skeletal/somatic features) without the biochemistry.

### BOARD TRAP

WRONG: ordering aggressive calcium workup/treatment for a PPHP patient — their labs are NORMAL. PPHP = the Albright LOOK with NORMAL chemistry; PHP1A = the Albright look WITH hypocalcemia/resistance. Same family, same mutation, opposite labs depending on parent of origin.

## 6. PHP1B (methylation)

### WHAT YOU SEE

Masked performer PHP1B: maternal GNAS methylation defect; PTH resistance only, no/mild AHO

### WHAT IT ENCODES

PHP type 1B: a maternal EPIGENETIC (methylation/imprinting) defect at the GNAS locus rather than a coding mutation. It causes renal PTH resistance (low Ca, high PO4, high PTH) and often mild TSH resistance, but typically LACKS the AHO body habitus and usually lacks the broad multi-hormone resistance seen in 1A.

### BOARD TRAP

WRONG: ruling out pseudohypoPTH because the patient looks physically normal (no brachydactyly/round face). PHP1B has the full biochemistry of resistance but NO AHO phenotype — the skeleton/somatic exam can be entirely normal, so you must rely on the labs (low Ca, high PO4, high PTH) and methylation testing, not the appearance.

## 7. Albright (AHO) phenotype

### WHAT YOU SEE

AHO Phenotype Ensemble: round/moon face, short stature, obesity, short 4th/5th metacarpals

### WHAT IT ENCODES

Albright hereditary osteodystrophy (AHO) is the somatic phenotype of Gs-alpha haploinsufficiency: SHORT STATURE, obesity, ROUND/moon face, and BRACHYDACTYLY — characteristically shortened 4th and 5th METACARPALS (a positive 'knuckle-knuckle-DIMPLE-dimple' sign when making a fist). Seen in both PHP1A (with resistance) and PPHP (without).

### BOARD TRAP

WRONG: equating AHO appearance with a specific diagnosis. AHO occurs in BOTH PHP1A and PPHP — appearance alone cannot tell you whether the patient has hormone resistance. You must add the labs: AHO + hypocalcemia/high PTH = PHP1A; AHO + normal labs = PPHP.



## 8. Ectopic / SC calcifications

### WHAT YOU SEE

AHO ensemble: heterotopic/subcutaneous ossifications, variable intellectual impairment

### WHAT IT ENCODES

Additional AHO features: SUBCUTANEOUS/heterotopic ossifications (ectopic bone formation in skin/soft tissue), basal ganglia calcifications, dental abnormalities, and variable intellectual disability/developmental delay. These reflect the broad role of Gs-alpha across tissues.

### BOARD TRAP

WRONG: mistaking the subcutaneous ossifications of AHO for the metastatic soft-tissue calcification of hyperphosphatemia. AHO ossifications are part of the genetic Gs-alpha phenotype (present in PPHP with normal Ca/PO4 too), not simply a product of high calcium-phosphate product — though chronic hyperphosphatemia in PHP1A can compound calcifications.

## 9. Ellsworth-Howard test

### WHAT YOU SEE

PHP Diagnostic Scroll: give exogenous PTH -> measure urinary cAMP / phosphate response

### WHAT IT ENCODES

The Ellsworth-Howard test confirms PTH resistance: infuse EXOGENOUS PTH and measure the urinary cAMP and phosphate (phosphaturic) response. In PHP1A/1B the kidney CANNOT respond -> BLUNTED rise in urinary cAMP and phosphate (because Gs-alpha can't make cAMP). In true hypoparathyroidism the kidney is normal -> a brisk, intact urinary cAMP/phosphate response.

### BOARD TRAP

WRONG: interpreting a blunted urinary cAMP response as 'no PTH effect, so the gland must be failing.' A blunted/absent response to GIVEN PTH proves end-organ RESISTANCE (pseudohypoPTH); an INTACT response to given PTH points to true hypoPTH (the kidney works, the problem is lack of endogenous PTH). The response direction is the discriminator.

## 10. PHP high vs hypoPTH low

### WHAT YOU SEE

Compare banner: PHP -> PTH HIGH / hypoPTH -> PTH LOW

### WHAT IT ENCODES

Board discriminator table: BOTH pseudohypoPTH and true hypoPTH share LOW calcium + HIGH phosphate. The separator is PTH — pseudohypoPTH has HIGH PTH (gland responding to resistance), true hypoPTH has LOW/inappropriately-normal PTH (gland failure or surgical removal). Renal function and 25-OH-D are normal in both (that excludes CKD and vitamin D deficiency).

### BOARD TRAP

WRONG: thinking the calcium/phosphate alone can separate the two — they're IDENTICAL (low Ca, high PO4). Only the PTH level (and the Ellsworth-Howard response) distinguishes them. High PTH with low Ca/high PO4 and normal kidneys/vitamin D = resistance = pseudohypoPTH.

## 11. Same mutation opposite biochem

### WHAT YOU SEE

Family Portrait Gallery: same mutation, OPPOSITE biochemistry — depends on parent of origin

### WHAT IT ENCODES

Within one family, the IDENTICAL GNAS loss-of-function allele produces opposite biochemistry across generations because of imprinting: when MOM transmits it, offspring get PHP1A (resistance + AHO); when DAD transmits it, offspring get PPHP (AHO only, normal labs). The somatic AHO phenotype passes either way; only the hormone resistance is parent-of-origin dependent.

### BOARD TRAP

WRONG: predicting a child's calcium status purely from the parent's diagnosis or the presence of the mutation. You must know which PARENT transmitted the allele — maternal transmission gives resistance (PHP1A), paternal gives no resistance (PPHP). This imprinting twist is a favorite exam point.

## 12. Acrodysostosis cousins (PRKAR1A)

### WHAT YOU SEE

Acrodysostosis Cousins: PRKAR1A or PDE4D/PDE3A defects mimic AHO downstream of GNAS

### WHAT IT ENCODES

Acrodysostosis and related disorders mimic the AHO skeletal phenotype (brachydactyly, short stature, facial features) but are caused by defects DOWNSTREAM of Gs-alpha in the cAMP/PKA pathway — PRKAR1A (regulatory subunit of protein kinase A) or PDE4D/PDE3A (phosphodiesterases). Some have hormone resistance, some don't.

### BOARD TRAP

WRONG: assuming every AHO-like child has a GNAS mutation. Genetically distinct mimics (PRKAR1A, PDE4D) act at later steps of the cAMP/PKA cascade and produce overlapping skeletal features — negative GNAS testing in an AHO-like patient should prompt consideration of these acrodysostosis 'cousins.'